EXP NO:5 Data Collection and Initial Exploration

AIM:
To collect a data set and perform initial exploration of the data.

PROGRAM:

python

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Data collection - reading the dataset
df = pd.read_csv(
    'https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv'
)
df

# First five rows
df.head()

# Last five rows
df.tail()

# Shape of the dataset (rows, columns)
df.shape

# Column names
df.columns

# Data types of each column
df.dtypes

# Information about the dataset
df.info()

# Statistical summary
df.describe()

# Check missing values
df.isnull().sum()

# Check duplicate rows
df.duplicated().sum()

# Unique values and counts of the species column
df['species'].unique()
df['species'].value_counts()

# Mean of each feature for every species
df.groupby('species').mean()

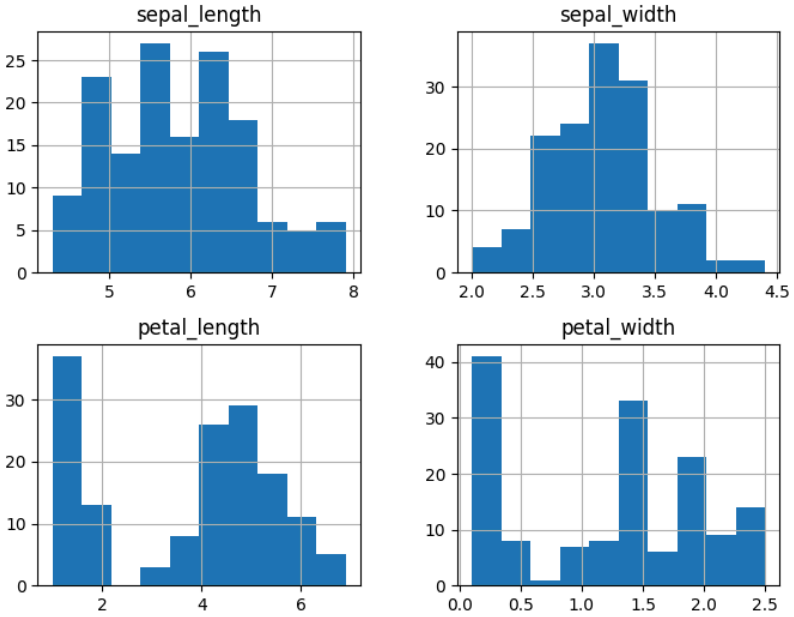
# Correlation between numerical features
df.drop(columns=['species']).corr()

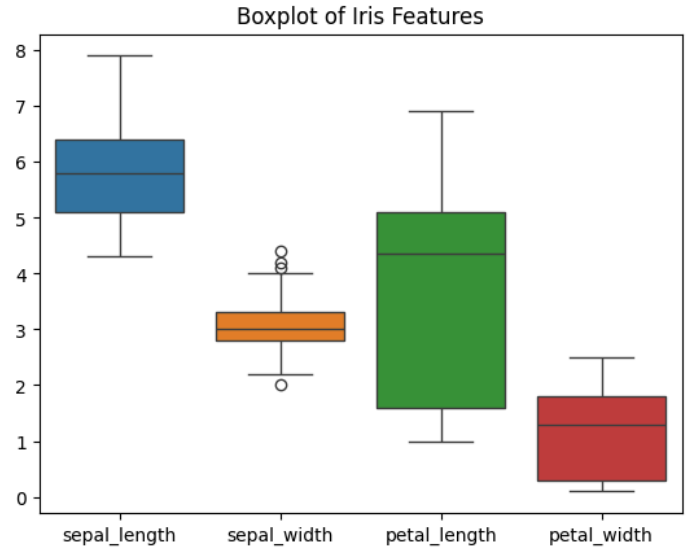
# Histogram of all numerical features
df.hist(figsize=(8, 6))
plt.show()

# Boxplot of the features
sns.boxplot(data=df.drop(columns=['species']))
plt.title("Boxplot of Iris Features")
plt.show()

# Pairplot
sns.pairplot(df, hue='species')
plt.show()
```

OUTPUT:





RESULT:

Thus the program to collect a data set and perform initial exploration of the data was executed successfully.